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CodeSwitch

Intelligent COBOL Refactoring

Voice Assistant

History 1

Gemini API Connected

Upload .cbl

Load Demo (3007 LOC)

sample.cbl

Export

Refactor with Gemini

Risk: 85 confidence

COBOL (Source)

1 IDENTIFICATION DIVISION.

2 PROGRAM-ID. PAYROLL01.

3 AUTHOR. GLOBAL-BANKING-LEGACY-1987.

4

5 * PAYROLL SYSTEM - GROSS/NET CALCULATION MODULE *

6 * WARNING: 1995 TAX RATES - OBSOLETE - REQUIRES UPDATE *

7

8

9 DATA DIVISION.

10 WORKING-STORAGE SECTION.

11

12 01 EMP-HOURLY-RATE PIC S9(5)V99 COMP-3.

13 01 EMP-STATUS PIC X(1).

14 88 EMP-ACTIVE VALUE 'A'.

15 88 EMP-TERMINATED VALUE 'T'.

16

17 * 1995 TAX RATES - NOT COMPLIANT WITH 2025 REGULATIONS

18 01 WS-TAX-BRACKETS-1995.

19 05 WS-BRACKET-1-LIMIT PIC 9(7) VALUE 23350.

20 05 WS-BRACKET-2-LIMIT PIC 9(7) VALUE 56550.

21 05 WS-RATE-BRACKET-1 PIC V999 VALUE .150.

22 05 WS-RATE-BRACKET-2 PIC V999 VALUE .280.

Python

Tests

Config

Diff

Architecture

Modules

NEW

1 python

2 import decimal

3

4 class Payroll:

5 def __init__(self, hourly_rate):

6 self.hourly_rate = decimal.Decimal(hourly_rate)

7 self.tax_brackets_1995 = {

8 'bracket_1_limit': decimal.Decimal('23350'),

9 'bracket_2_limit': decimal.Decimal('56550'),

10 'rate_bracket_1': decimal.Decimal('0.150'),

11 'rate_bracket_2': decimal.Decimal('0.280')

12 }

13 self.fica_rates = {

14 'ss_rate': decimal.Decimal('0.0620'),

15 'medicare_rate': decimal.Decimal('0.0145'),

16 'ss_wage_base': decimal.Decimal('61200')

17 }

18 self.gross_pay = decimal.Decimal('0.00')

19 self.federal_tax = decimal.Decimal('0.00')

20 self.fica_tax = decimal.Decimal('0.00')

21 self.net_pay = decimal.Decimal('0.00')

22

Transformation Metrics

LIVE

58

COBOL

→

52

Python

(business code)

6

Tests

(unit tests)

52

Total

(delivered)

2

Issues

5

Improvements

85%

Confidence

(validated)

Test Oracle - Equivalence Validation

PASSED

6

Tests Generated

6

Tests Passed

0

Tests Failed

100%

Coverage

COBOL ↔ Python Equivalence: All test cases validated successfully

Tested 6 scenarios including edge cases, boundary conditions, and error handling.

Gemini Live Chat

ONLINE

U

what is cocol

G

"Cocol" most commonly refers to a type of **sweet bread or roll.** often found in Mexican cuisine.

Migration Summary

This COBOL program, PAYROLL01, calculates an employee's gross pay, federal tax, FICA tax, and net pay based on an hourly rate. It uses hardcoded tax rates and FICA limits from 1995, which are outdated and need to be updated for current compliance. The program's primary function is payroll calculation, but it utilizes obsolete tax information.

Complexity

Medium

Risk Level

Estimated Effort

Full cycle

Confidence

85

Ready for UAT

